Example 1. Carried over from yesterday $f(x) = x\sqrt{4-x^2}$ on $[-2, 2]$

Given: $f'(x) = \dfrac{2\left(2-x^2\right)}{\sqrt{4-x^2}}$ $\qquad$ $f''(x) = \dfrac{2x\left(x^2-6\right)}{\left(4-x^2\right)^{3/2}}$

1. Domain $4 - x^2 \geq 0 \Rightarrow$ **[−2, 2]** — closed and bounded

2. Intercepts $(0, 0)$ and $(\pm 2, 0)$

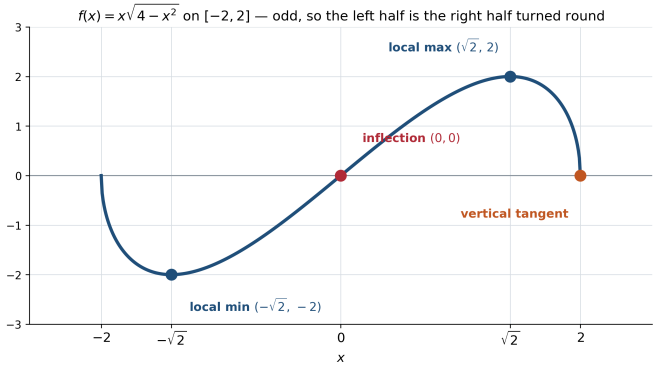
3. Symmetry $f(-x) = -f(x)$, so **odd** — do the right half and turn it round

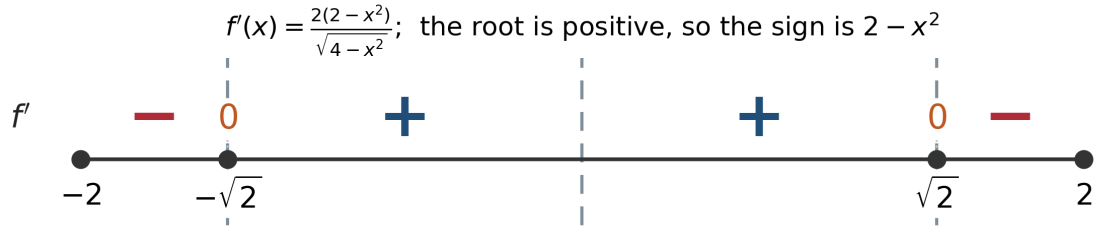
4. Asymptotes **none** — the domain stops at ± 2, there is no infinity to run to

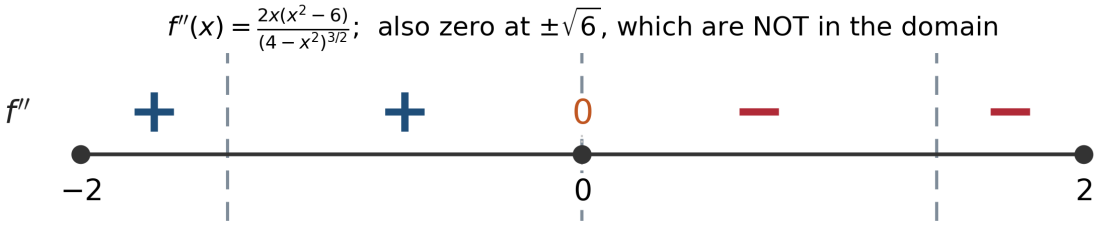
5. Increase / decrease incr. on $(-\sqrt{2}, \sqrt{2})$; decr. on $(-2, -\sqrt{2})$ and $(\sqrt{2}, 2)$

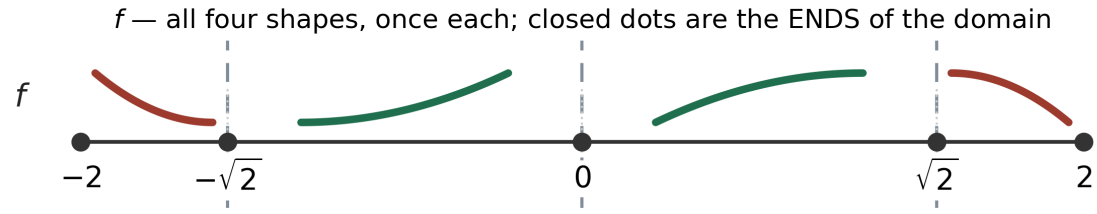
6. Local extrema min $(-\sqrt{2}, -2)$, max $(\sqrt{2}, 2)$

7. Concavity / inflection CU on $(-2, 0)$, CD on $(0, 2)$; **one** inflection point $(0, 0)$









$f'' = 0$ at 0 *and* $\pm\sqrt{6} \approx \pm 2.449$ — but $\pm\sqrt{6}$ are **not in the domain**. And $f' \to -\infty$ at $x = \pm 2$: the curve meets the axis **vertically**, which is what stops the sketch looking like a cubic.

Example 2. An asymptote on one side only $f(x) = x^2 e^{-x}$

Given: $f'(x) = x(2-x)e^{-x}$ $f''(x) = \left(x^2 - 4x + 2\right) e^{-x}$

1. Domain all real numbers

2. Intercepts $(0, 0)$ only — an exponential is never zero

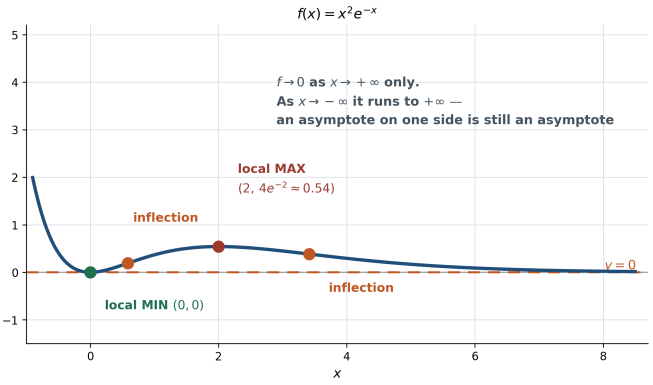
3. Symmetry neither

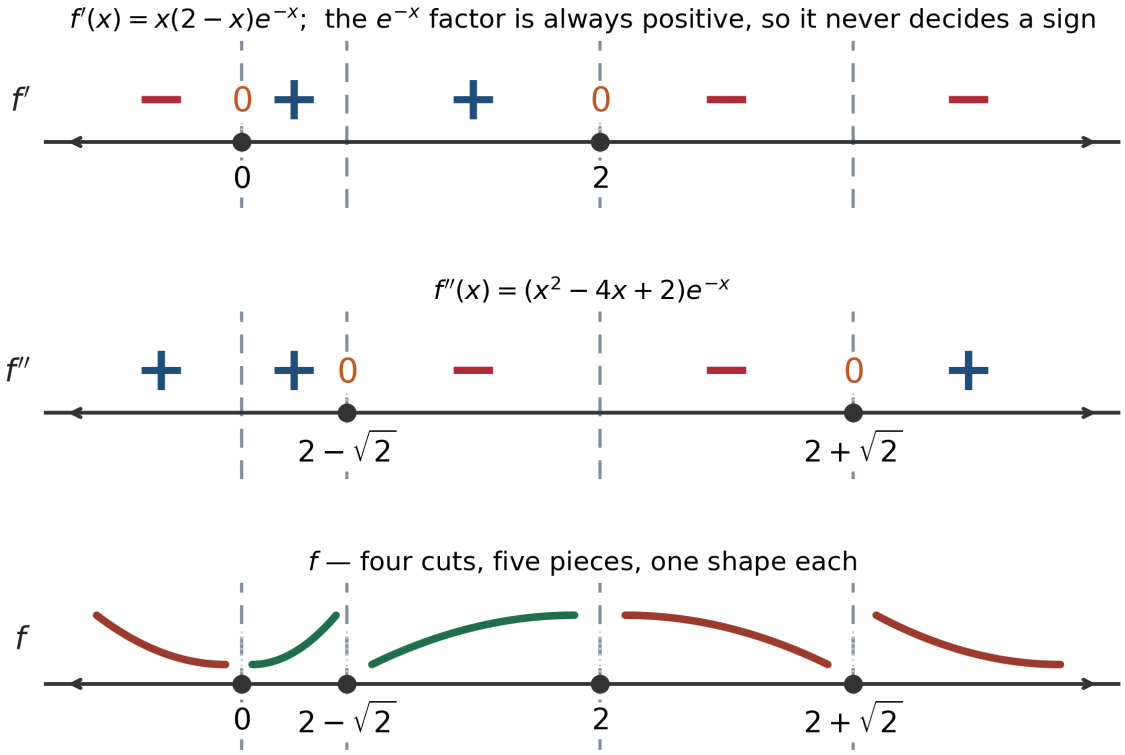
4. Asymptotes $y = 0$ **to the right only**; as $x \to -\infty$, $f \to +\infty$

5. Increase / decrease decr. on $(-\infty, 0)$; incr. on $(0, 2)$; decr. on $(2, \infty)$

6. Local extrema min $(0, 0)$, max $\left(2,\, 4e^{-2}\right) \approx (2,\, 0.54)$

7. Concavity / inflection CU on $(-\infty, 2 - \sqrt{2})$ and $(2+\sqrt{2}, \infty)$, CD between; inflections at $x = 2 \pm \sqrt{2}$





The two ends are **separate questions**. e^{-x} blows up as $x \to -\infty$ — the minus sign does not mean decay in both directions.

Example 3. Exactly what Exam 3 asks $f(x) = \dfrac{x^2-4}{x^2+4}$

Given: $f'(x) = \dfrac{16x}{(x^2+4)^2}$ $\qquad f''(x) = \dfrac{-16\left(3x^2-4\right)}{(x^2+4)^3}$

1. Domain all real numbers — x^2+4 is never zero, so **no vertical asymptotes**

2. Intercepts $(0,-1)$ and $(\pm 2,\, 0)$

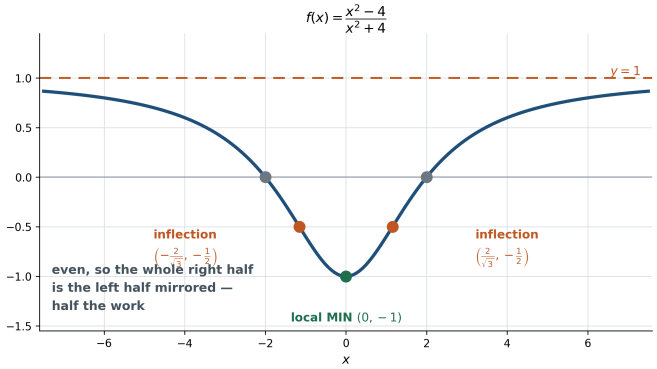
3. Symmetry every power is even, so **even**

4. Asymptotes $y = 1$ in both directions; the curve stays *below* it throughout

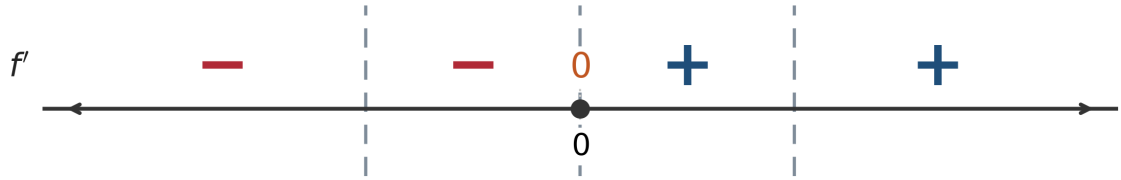
5. Increase / decrease decr. on $(-\infty, 0)$; incr. on $(0, \infty)$

6. Local extrema min $(0,-1)$ — and by the symmetry it is the **absolute** minimum

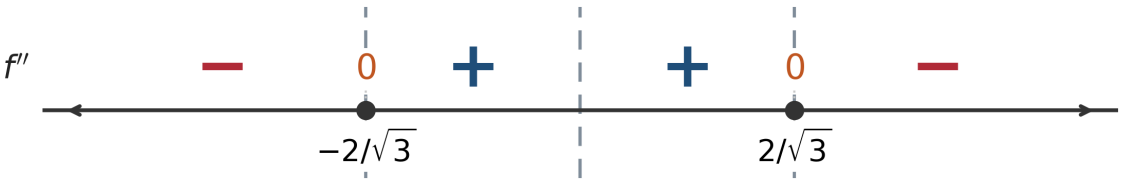
7. Concavity / inflection CU on $\left(-\frac{2}{\sqrt{3}}, \frac{2}{\sqrt{3}}\right)$, CD outside; inflections $\left(\pm\frac{2}{\sqrt{3}}, -\frac{1}{2}\right)$



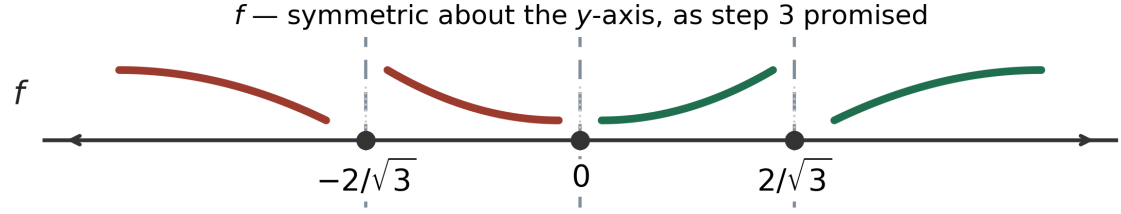
$f'(x) = 16x/(x^2+4)^2$; denominator positive, so the sign is $16x$



$f''(x) = 16(4-3x^2)/(x^2+4)^3$



f — symmetric about the y-axis, as step 3 promised



This is Exam 3's shape: f, f' and f'' handed over, four lettered parts, then a sketch. The shape line *is* the sketch.

Example 4. A fractional exponent, and a cusp $f(x) = x^{\frac{2}{3}}(x-5)$

Given: $f'(x) = \dfrac{5(x-2)}{3x^{1/3}}$ $\qquad$ $f''(x) = \dfrac{10(x+1)}{9x^{4/3}}$

1. Domain all real numbers — a cube root accepts negatives

2. Intercepts $(0,0)$ and $(5,0)$

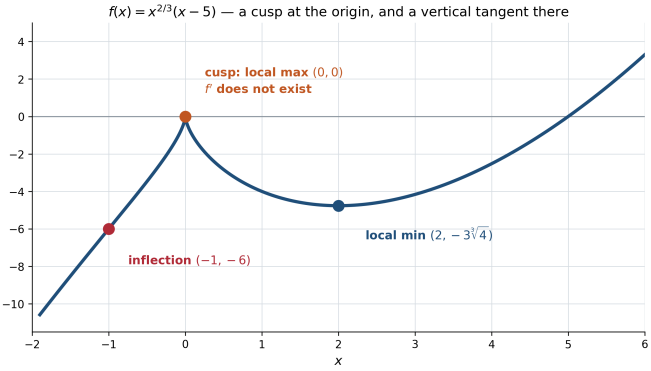
3. Symmetry neither

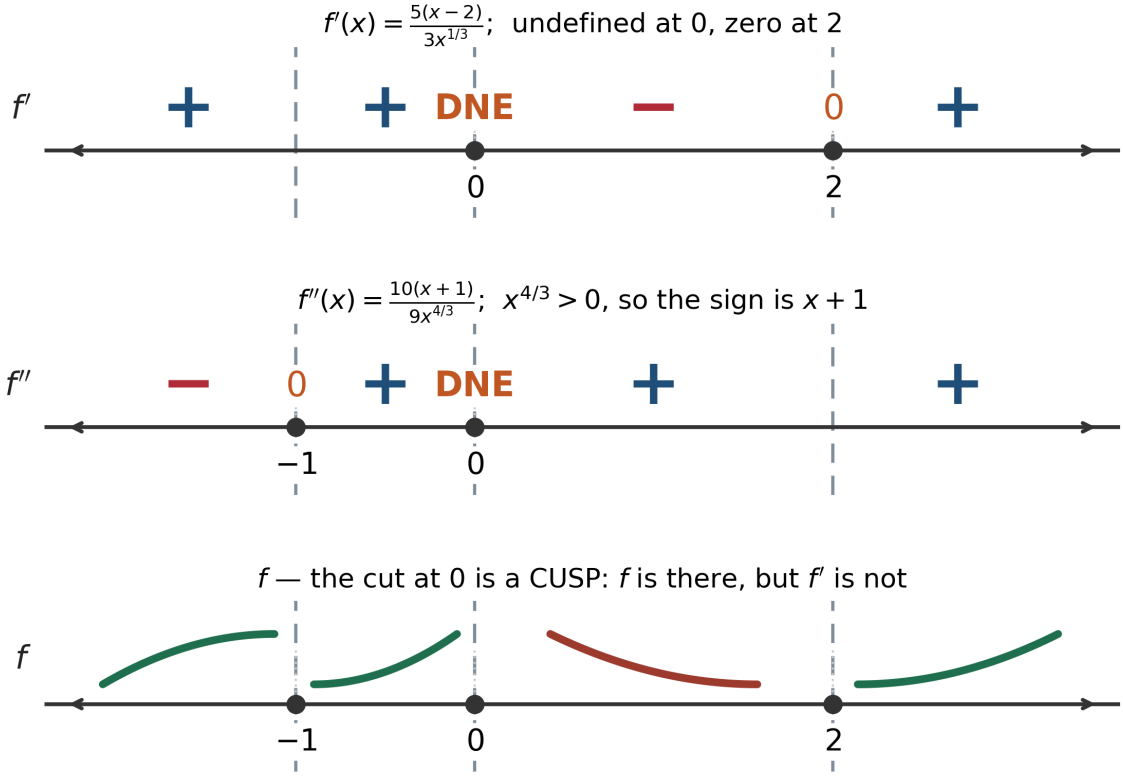
4. Asymptotes none

5. Increase / decrease incr. on $(-\infty, 0)$; decr. on $(0,2)$; incr. on $(2,\infty)$

6. Local extrema **cusp** local max $(0,0)$; local min $\left(2, -3\sqrt[3]{4}\right) \approx (2, -4.76)$

7. Concavity / inflection CD on $(-\infty,-1)$, CU on $(-1,0)$ and $(0,\infty)$; inflection $(-1,-6)$





$x = 0$ is a **critical number** even though $f'(0)$ does not exist — the definition is "$f'(c) = 0$ *or* $f'(c)$ undefined". And $x^{4/3} > 0$ on both sides, so the concavity does *not* change at 0.

Example 5. A trigonometric one $f(x) = x + 2\cos(x)$ on $[0, 2\pi]$

Given: $f'(x) = 1 - 2\sin x$ $\quad$ $f''(x) = -2\cos x$

1. Domain given as $[0, 2\pi]$ — **closed**, so the endpoints are candidates

2. Intercepts $g(0) = 2$; **no** x-intercepts (the smallest value is ≈ 0.89)

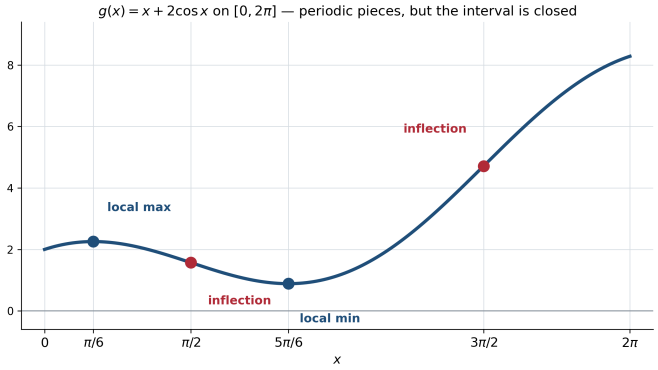
3. Symmetry neither — and on a restricted interval symmetry buys nothing anyway

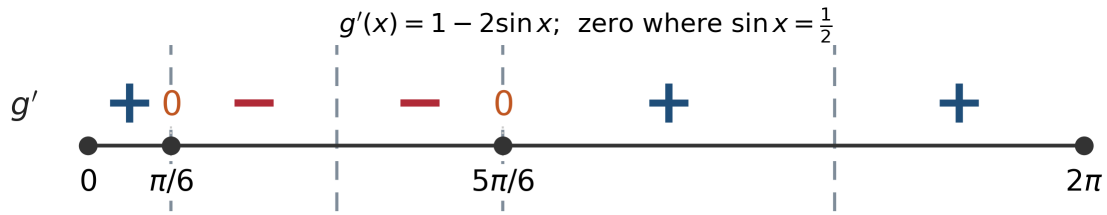
4. Asymptotes none on a closed interval

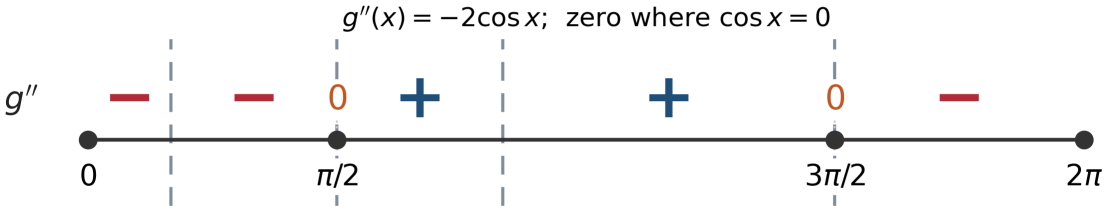
5. Increase / decrease incr. on $\left(0, \frac{\pi}{6}\right)$; decr. on $\left(\frac{\pi}{6}, \frac{5\pi}{6}\right)$; incr. on $\left(\frac{5\pi}{6}, 2\pi\right)$

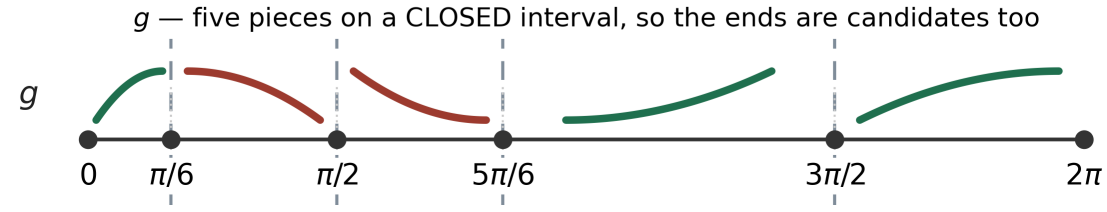
6. Local extrema max $\left(\frac{\pi}{6}, \frac{\pi}{6} + \sqrt{3}\right) \approx 2.26$; min $\left(\frac{5\pi}{6}, \frac{5\pi}{6} - \sqrt{3}\right) \approx 0.89$

7. Concavity / inflection CD on $\left(0, \frac{\pi}{2}\right)$, CU on $\left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$, CD on $\left(\frac{3\pi}{2}, 2\pi\right)$; inflections at $\frac{\pi}{2}$ and $\frac{3\pi}{2}$









The absolute maximum is an endpoint: $g(2\pi) = 2\pi + 2 \approx 8.28$ beats the local max at $\frac{\pi}{6}$. The absolute minimum is the local one at $\frac{5\pi}{6}$. Critical numbers came from *solving* $\sin x = \frac{1}{2}$, not from factoring.